PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Gender differences in faculty rank among academic physicians: A systematic review and meta-analysis
AUTHORS	Li, Ben; Jacob-Brassard, Jean; Dossa, Fahima; Salata, Konrad; Kishibe, Teruko; Greco, Elisa; Baxter, Nancy; Al-Omran, Mohammed

VERSION 1 – REVIEW

REVIEWER	Gandré, Coralie IRDES
REVIEW RETURNED	18-Apr-2021

GENERAL COMMENTS	Thank you for the opportunity to review this interesting paper which may contribute to raise awareness regarding persisting gender inequities in faculty rank among academic physicians worldwide. The paper is well-written and structured and as a result, very pleasant to read. I also cannot imagine the amount of time which has been necessary to select, read and assess the quality of all included research and I would like to acknowledge it. However, I still have some comments which may contribute to improve the paper overall. They are presented below (mostly by chronological order of appearance in the paper). 1/ In the abstract, it would be important to precise the geographical scope of your review (i.e., worldwide). 2/ In the abstract or at least in the paper, when you mention the number of academic physician data points, it would be important to add what it generally includes (professors, assistant professors, etc.). 3/ The introduction is very short, the problematic under study and its interest should probably be more developed. Readers need to understand better the processes of appointments to academic positions and how this may differ or not across settings. Also in the last paragraph, I find it disturbing that you focus on a few specialties and regions as you are supposed to explain the rationale for developing a systematic review and we wonder why you highlight a few studies out of the many existing. 4/ Statistical analysis: I find it disturbing (throughout the paper) that some of your secondary outcomes (such as publications and h-index in particular) are a proxy of productivity and are used in some studies as explanatory variables for your primary outcome (full professorship appointment). I would advise against using

	them as secondary outcomes. Also, their choice should be presented in a dedicated part on outcomes not in the statistical analysis. Still regarding outcomes, what is federal research funding? I imagine it is very specific to North America (or at least federal contexts), why did not you consider the obtention of research grants overall? This has been considered by some studies. 5/ Can you add a reference for Review Manager 5.4.1 as you do for the R software and packages used? 6/ Study characteristics: give more details on the academic physician data points included (which types of positions/appointments?); a table or figure of the number of included studies by country would also be helpful (also as your paper is a little bit North American-centric in some parts). 7/ Baseline demographic data: this part is confusing without any info on the academic ranks included here. 8/ Meta-analysis of full-professorship: Is 1979 the year for the first study available on the topic of interest? Mention it clearly. Also present the data on subgroup analysis by decade by chronological order so readers can clearly follow the evolution over time (currently you first presented the most recent decade). 9/ I am not fully convinced by the grouping of all research outside of North America altogether in comparison to North America. On some aspects, the North American context might be closer to the European one than the Asian one for instance (or the contrary but we need to know the rationale behind). This choice should be more strongly justified. 10/ In the sub-analyses by specialties, how did you deal with the potential heterogeneity of what is included in different specialties according to national contexts? 11/ Meta-analyses of secondary outcomes: Salaries are presented in dollars, is it US or Canadian dollars? Also how did you convert all salaries to these dollars (for studies outside of the US or Canadian context)? 12/ Discussion, comparison to the existing literature: once again like in the introduction, you focus on a few random studies, this is disturbing. Why did you pick those specific studies out? Are they the ones with the best quality assessment? 13/ The examples of interventions in the discussion are also very North American centric. I wonder if you could also find one outside of this context? 14/ Limitations: “that did not statistically adjusting for confounding”: I think you meant “adjust” not “adjusting”.

REVIEWER	Ellinas , Elizabeth Medical College of Wisconsin, Center for the Advancement of Women in Science and Medicine
REVIEW RETURNED	17-May-2021

GENERAL COMMENTS	Gender differences in faculty rank among academic physicians: A systematic review and meta-analysis I thank the authors for their review and meta-analysis of faculty rank by gender in academic medicine worldwide. Controlling for multiple factors in a multi-national sample, they found that men were 2.77 times more likely to be full professors than women. This is important work involving hundreds of thousands of faculty. The sample here is, of course, everything. I'm curious as to what "original studies reporting faculty rank" means? Many studies report rank of participants. More information would be helpful regarding what sort of studies were accepted. And I see that you provide the entire list. Some description within the text of your work would be helpful. Further, how did the authors guard against counting people more than once? For example, an anesthesiologist in the US could be counted in a study by their institution, by a study of US anesthesiologists, and by the AAMC. For me, meta-analysis of the secondary outcomes was at least, if not more interesting. This includes, for example, that funnel plots suggesting publication bias of studies with greater gender differences in several areas, including publication and salary differences. Figure 2&5: these forest plots are very small and unreadable for me. Figure 4: Please provide more description regarding what this is – Y axis needs to share log odds of what, although I assume it's OR women and men professors?

REVIEWER	Dadi, A.F. Menzies School of Health Research, Centre for Child Development and Education
REVIEW RETURNED	23-Jul-2021

GENERAL COMMENTS	Thank you reviewer for conducting this study. I am asked to review the data analysis section of the methodology. I have just one concern probably suggesting different analysis (time trend analysis) approaches to see the trend over time rather than using meta-regression.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Coralie Gandré, IRDES

Thank you for the opportunity to review this interesting paper which may contribute to raise awareness regarding persisting gender inequities in faculty rank among academic physicians worldwide. The paper is well-written and structured and as a result, very pleasant to read. I also cannot imagine the amount of time which has been necessary to select, read and assess the quality of all included research and I would like to acknowledge it. However, I still have some comments which may contribute to improve the paper

overall. They are presented below (mostly by chronological order of appearance in the paper).

1/ In the abstract, it would be important to precise the geographical scope of your review (i.e., worldwide).

Response: We have updated the study selection section of our abstract to the following: “All original studies reporting faculty rank stratified by gender worldwide were included.” (Page 2, lines 33-34).

2/ In the abstract or at least in the paper, when you mention the number of academic physician data points, it would be important to add what it generally includes (professors, assistant professors, etc.).

Response: We could not determine the academic ranks of each participant in our cohort (e.g., instructors and assistant/associate professors) as many studies only reported the number of full professors from a pool of academic physicians. However, we have updated our methods section with the following sentence: “An academic physician was defined as a physician with a medical school appointment (e.g., instructors and assistant/associate/full professors) (38).” (Page 8, lines 165-166). Furthermore, we have included the following statement in our limitations section: “In our cohort, we did not have information on the academic ranks of all the participants (e.g., instructors and assistant/associate professors) as many studies only reported the number of full professors out of a pool of academic physicians. We recommend that future studies on this topic report the academic ranks of all their participants.” (Page 20, lines 444-447).

3/ The introduction is very short, the problematic under study and its interest should probably be more developed. Readers need to understand better the processes of appointments to academic positions and how this may differ or not across settings. Also in the last paragraph, I find it disturbing that you focus on a few specialties and regions as you are supposed to explain the rationale for developing a systematic review and we wonder why you highlight a few studies out of the many existing.

Response: We have lengthened the introduction and expanded on the problem under study. Furthermore, we added a paragraph describing the processes of academic rank appointments and how this differs across settings. The last paragraph has been revised and we have removed mention of specific specialties and regions. The introduction now reads as follows.

“Gender equity remains an important issue in academic medicine. In the United States, women make up more than half of medical school graduates (1) and 37% of full-time faculty (2). However, only 24% of full professors (3), 18% of department chairs (4), and 17% of deans are women (5). Women are under-represented on editorial boards, have fewer publications, and receive less research funding (6,7). Due to these inequities, women are more likely to leave academia (8). Previous studies have demonstrated the positive impact of health care workforce diversity on patient outcomes, costs, and innovation (9). Therefore, gender imbalances in academic medicine, particularly in leadership positions, have broad negative implications (10).

An important indicator of gender inequity in academic medicine is the difference in the proportion of women vs. men with full professorship (11). The decision to appoint a physician to full professor is usually made by a medical school committee and generally takes into consideration a candidate’s clinical and academic contributions (12). However, this varies across settings based on institutional values, specialty, and conscious or unconscious bias of promotions committee members (13–15). Importantly, several studies have demonstrated that women are less likely than men to be full professors even after adjusting for clinical and academic productivity (16–18).

The current literature on gender bias in academic medicine is limited by single studies that assess one institution (23,24), specialty (25–28), or country (29–33). Furthermore, the cross-sectional nature of most analyses precludes an investigation of trends over time (34–36). We conducted a systematic review and meta-analysis to calculate a pooled measurement of gender differences in full professorship worldwide and assessed variations across specialties, between geographic regions, and over time." (Page 5-6, lines 94-124).

4/ Statistical analysis: I find it disturbing (throughout the paper) that some of your secondary outcomes (such as publications and h-index in particular) are a proxy of productivity and are used in some studies as explanatory variables for your primary outcome (full professorship appointment). I would advise against using them as secondary outcomes. Also, their choice should be presented in a dedicated part on outcomes not in the statistical analysis. Still regarding outcomes, what is federal research funding? I imagine it is very specific to North America (or at least federal contexts), why did not you consider the obtention of research grants overall? This has been considered by some studies.

Response: We agree that our secondary outcomes publications and h-index are measurements of research productivity and sometimes used as explanatory variables for gender differences in academic rank. The reason we chose to include these secondary outcomes were to provide readers with an understanding of the magnitude of difference in these variables between women and men. This is important because in our pooled analysis of adjusted outcomes, several studies control for publications and h-index. By seeing the gender-based effect size difference, readers can better understand the potential impact of research productivity measurements on full professorship status. Furthermore, publications and h-index can be viewed as measurements of academic success independent of their contributions to academic rank (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4987709/#!po=14.5161). Another reviewer of this manuscript commented they found the results of the meta-analysis of our secondary outcomes interesting, including publications. For these reasons, we would like to continue including publications and h-index as secondary outcomes.

Federal research funding was defined as any funding received by academic physicians from a national government. Although common in North America, examples of federal research funding agencies exist in other settings, including the UK Research Councils, French National Research Agency, and National Natural Science Foundation of China. The reason we chose federal research funding as a secondary outcome was because the recipients of these types of funding are highly selected and therefore an important indicator of academic productivity (https://pubmed.ncbi.nlm.nih.gov/26818308/). We agree that gender differences in overall research grant funding would be important to assess, however, our search yielded only 5 studies that reported this outcome. Therefore, there was not sufficient data for a meta-analysis of overall research grants.

We have presented our choices of primary and secondary outcomes in a dedicated section in our methods, which reads as follows.

"The primary outcome was the difference in the proportion of female vs. male academic physicians with full professorship. An academic physician was defined as a physician with a medical school appointment (e.g., instructors and assistant/associate/full professors) (38). This primary outcome was chosen because full professorship is generally considered the highest academic rank and an important indicator of the overall status of an academic physician (11).

The secondary outcomes were total publications, h-index, salary, proportion receiving federal research funding, and leadership (proportion who are departmental chairs). Total publications, h-index, and federal

research funding are markers of academic productivity (39). Federal research funding was defined as any funding received by academic physicians from a national government (40). Salary and departmental chair positions are other indicators of physician status (41,42)." (Page 8, lines 162-173).

5/ Can you add a reference for Review Manager 5.4.1 as you do for the R software and packages used?

Response: Added reference 49 (page 9, line 197).

6/ Study characteristics: give more details on the academic physician data points included (which types of positions/appointments?); a table or figure of the number of included studies by country would also be helpful (also as your paper is a little bit North American-centric in some parts).

4/ Statistical analysis: I find it disturbing (throughout the paper) that some of your secondary outcomes (such as publications and h-index in particular) are a proxy of productivity and are used in some studies as explanatory variables for your primary outcome (full professorship appointment). I would advise against using them as secondary outcomes. Also, their choice should be presented in a dedicated part on outcomes not in the statistical analysis. Still regarding outcomes, what is federal research funding? I imagine it is very specific to North America (or at least federal contexts), why did not you consider the obtention of research grants overall? This has been considered by some studies.

Response: We agree that our secondary outcomes publications and h-index are measurements of research productivity and sometimes used as explanatory variables for gender differences in academic rank. The reason we chose to include these secondary outcomes were to provide readers with an understanding of the magnitude of difference in these variables between women and men. This is important because in our pooled analysis of adjusted outcomes, several studies control for publications and h-index. By seeing the gender-based effect size difference, readers can better understand the potential impact of research productivity measurements on full professorship status. Furthermore, publications and h-index can be viewed as measurements of academic success independent of their contributions to academic rank (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4987709/#!po=14.5161). Another reviewer of this manuscript commented they found the results of the meta-analysis of our secondary outcomes interesting, including publications. For these reasons, we would like to continue including publications and h-index as secondary outcomes.

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full professorship is generally considered the highest academic rank and an important indicator of the overall status of an academic physician (11).

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Response: Added reference 49 (page 9, line 197).

6/ Study characteristics: give more details on the academic physician data points included (which types of positions/appointments?); a table or figure of the number of included studies by country would also be helpful (also as your paper is a little bit North American-centric in some parts).

Response: We do not have information on the positions/appointments of all the participants in our cohort (e.g., instructors and assistant/associate professors) as many studies only reported the number of full professors out of a pool of academic physicians. We have included the following statement in our study characteristics section: "The population included instructors and assistant/associate/full professors." (Page 10, lines 223-224). Furthermore, we have included the following statement in our limitations section: "In our cohort, we did not have information on the academic ranks of all the participants (e.g., instructors and assistant/associate professors) as many studies only reported the number of full professors out of a pool of academic physicians. We recommend that future studies on this topic report the academic ranks of all their participants." (Page 20, lines 444-447).

We have added eTable 2b (Supplement, page 20) detailing the number of included studies by country of publication and added the following statement to our study characteristics section: "Outcomes were reported on 24 specialties across 16 regions, including countries from Europe (7), Asia (7), and North America (2)." (Page 10, lines 229-230).

7/ Baseline demographic data: this part is confusing without any info on the academic ranks included here.

Response: We agree that it would be helpful for the reader to know the academic ranks of all the included participants and recognize this lack of information as a limitation of our study. We have included the following statement in our limitations section: "In our cohort, we did not have information on the academic ranks of all the participants (e.g., instructors and assistant/associate professors) as many studies only reported the number of full professors out of a pool of academic physicians. We recommend that future studies on this topic report the academic ranks of all their participants." (Page 20, lines 444-447).

8/ Meta-analysis of full-professorship: Is 1979 the year for the first study available on the topic of interest? Mention it clearly. Also present the data on subgroup analysis by decade by chronological order so readers can clearly follow the evolution over time (currently you first presented the most recent decade).

Response: Yes, 1979 was the data collection year for the first study available on this topic. We have added the following statement to our meta-analysis of full professorship section: "Of note, 1979 was the

data collection year for the first study on gender differences in faculty rank of academic physicians based on our search." (Page 12, lines 271-272).

We have updated the text on subgroup analysis by decade to now be in chronological order (Before 2000 > 2000-2009 > 2010-2020, page 12-13, lines 272-277).

9/ I am not fully convinced by the grouping of all research outside of North America altogether in comparison to North America. On some aspects, the North American context might be closer to the European one than the Asian one for instance (or the contrary but we need to know the rationale behind). This choice should be more strongly justified.

Response: We agree that ideally, we would be able to analyze the data on a country-level basis. However, the majority of studies on this topic were published in North America (182/218). Most countries in Europe and Asia had 5 or fewer studies. Therefore, there was not sufficient data to perform a subgroup meta-analysis by country. We have included the following statement in our results section: "Given that the majority of included studies were published in North America (182/218), we compared countries within North America to countries outside of North America." (Page 13, lines 278-280). As additional data from non-North American regions accumulate, we agree that it would be prudent to assess gender differences in faculty rank on a country-level basis.

10/ In the sub-analyses by specialties, how did you deal with the potential heterogeneity of what is included in different specialties according to national contexts?

Response: We recognize the high heterogeneity of the pooled analysis by specialty and acknowledge this in our limitations section: "This study was limited by high heterogeneity in most of the analyses conducted. This was expected as we pooled effect estimates across multiple specialties, data sources, and countries." (Page 19, lines 426-427). Interestingly, subgroup analysis by specialty demonstrated a reduction in heterogeneity (e.g., I2 = 0% for cardiology, emergency medicine, hematology, vascular surgery, respirology, and urology). Therefore, data variability across specialties may be an important source of heterogeneity in our study. We have included a statement on this in our limitations section: "Subgroup analysis by specialty demonstrated a reduction in heterogeneity. Therefore, data variability across specialties may be a source of heterogeneity in our study." (Page 19, lines 428-429).

One method we used to deal with heterogeneity was performing sensitivity analysis removing poor quality studies and this did not have a significant impact on the pooled estimate for any outcome. We have mentioned this strategy in our limitations section: "Sensitivity analysis was also conducted to remove poor quality studies, which did not have a significant impact on the pooled estimate for any outcome." (Page 20, lines 437-438).

11/ Meta-analyses of secondary outcomes: Salaries are presented in dollars, is it US or Canadian dollars? Also how did you convert all salaries to these dollars (for studies outside of the US or Canadian context)?

Response: Salaries are presented in US dollars. We converted all salaries to US dollars using a standardized currency converter (https://www.xe.com/currencyconverter/). We have added the following statement to our methods section: "Monetary values for salary were converted to US dollars using a standardized currency converter (48)." (Page 9, lines 193-194).

12/ Discussion, comparison to the existing literature: once again like in the introduction, you focus on a few random studies, this is disturbing. Why did you pick those specific studies out? Are they the ones with the best quality assessment?

Response: We compared our findings to Jena et. al (2015) because they performed the largest single study on this topic, analyzing 91,073 US academic physicians (https://jamanetwork.com/journals/jama/fullarticle/2441260). We have removed mention of other single studies on specific specialties and countries. The comparison to existing literature section now reads as follows.

“This was the first systematic review and meta-analysis of gender differences in faculty rank among academic physicians. Our findings are consistent with most of the single studies in the literature demonstrating significant gender disparities in academic rank. Specifically, our pooled analysis of 24 specialties across 16 countries confirmed that the gender disparity in full professorship is pervasive across specialties globally.

Jena et al. (2015) performed the largest single study on this topic, analyzing 91,073 US academic physicians. They showed that 28.6% of men were full professors compared to 11.9% of women, for an absolute difference of 16.7% (16). In their study, men were older, more likely to hold federal research grants, and published more papers. However, after adjusting for experience, research productivity, and specialty, women continued to be less likely than men to be full professors, with an absolute difference of 4% (16). We demonstrated similar findings across specialties, over time, and between geographic regions in a larger pooled cohort.” (Page 16-17, lines 358-376).

13/ The examples of interventions in the discussion are also very North American centric. I wonder if you could also find one outside of this context?

Response: We have added the example of Athena SWAN (Scientific Women’s Academic Network) action plans applied to the European context to our discussion section: “In Europe, many medical schools have instituted Athena SWAN (Scientific Women’s Academic Network) action plans, which are a set of multi-level interventions that address structural, cultural, and institutional factors that create barriers for women (72). This broad-based strategy has significantly improved female representation in leadership positions (73).” (Page 19, lines 416-420).

14/ Limitations: “that did not statistically adjusting for confounding”: I think you meant “adjust” not “adjusting”.

Response: This error has been corrected (page 19, line 435).

Reviewer: 2
Dr. Elizabeth Ellinas, Medical College of Wisconsin

Response: We performed a broad search that aimed to capture all studies within 5 major databases that included the following keywords or synonyms in their text “gender AND physician AND faculty AND rank.” This returned 5,897 unique articles. From this pool, we included all studies that reported faculty rank stratified by gender. The studies did not need to be specifically designed to assess gender differences in faculty rank. As long as their study population included academic physicians and they reported the proportion of full professors in women and men, the article was included. This approach allowed us to

capture articles broadly and sensitively. We have added the following clarifying statements to our methods section: “Included articles did not need to be specifically designed to assess gender differences in faculty rank. As long as studies reported the proportion of full professors in a pool of academic physicians for women and men, they were included. This approach allowed us to capture articles broadly and sensitively.” (Page 7, lines 146-149).

Further, how did the authors guard against counting people more than once? For example, an anesthesiologist in the US could be counted in a study by their institution, by a study of US anesthesiologists, and by the AAMC.

Response: One strategy we used to reduce the risk of double counting participants was when we found that multiple studies were conducted using the same data source, year of data collection, and outcome, only the most comprehensive article was included (defined as largest sample size and/or most outcomes reported). However, as you correctly point out, there is still the potential to count a physician more than once if they were included in different data sources. This risk cannot be completely mitigated without participant-level data, which was not available for our meta-analysis. We have acknowledged this in our limitations section: “We intended to reduce the risk of counting a study participant more than once by including only the most comprehensive article (largest sample size and/or most outcomes reported) when multiple studies reported duplicate data (same data source, year of data collection, and outcome). However, this risk could not be completely mitigated without participant-level data, which was not available in our study-level meta-analysis.” (Page 20, lines 439-443).

For me, meta-analysis of the secondary outcomes was at least, if not more interesting. This includes, for example, that funnel plots suggesting publication bias of studies with greater gender differences in several areas, including publication and salary differences.

Response: Thank you for your comments.

Figure 2&5: these forest plots are very small and unreadable for me.

Response: The forest plots in Figures 2 and 5 have now been enlarged.

Figure 4: Please provide more description regarding what this is – Y axis needs to share log odds of what, although I assume it’s OR women and men professors?

Response: The legend of Figure 4 has been updated with the following sentence: “The Y axis represents the log odds ratio of the proportion of men vs. women with full professorship.” (Page 36, lines 822-823).

Reviewer: 3
Dr. A.F. Dadi, Menzies School of Health Research, The university of Gondar, College of Medicine and Health Sciences

Response: Thank you for your comments. We used a meta-regression approach to assess trends over time because this has been previously validated (https://pubmed.ncbi.nlm.nih.gov/19345563/) and commonly used in the literature (see 3 studies below).

VERSION 2 – REVIEW

REVIEWER	Ellinas , Elizabeth Medical College of Wisconsin, Center for the Advancement of Women in Science and Medicine
REVIEW RETURNED	01-Sep-2021

GENERAL COMMENTS	This is truly a massize undertaking and I commend the authors on their work. I have read the authors responses to all three reviewers and find their responses sufficient, with the exception of readability of the forest plots, which i trust can be remedied in publication. I have no further comments and look forward to seeing the finished paper.

REVIEWER	Dadi, A.F. Menzies School of Health Research, Centre for Child Development and Education
REVIEW RETURNED	17-Aug-2021

GENERAL COMMENTS	My concern has been resolved, thanks.